# MATH 464 Midterm #1

TOPICS: All lecture and reading material covered during the first $4\frac{1}{2}$ weeks of class. This includes:

- Modeling of MIP's from "word problems"
- Identifying MIP's, LP's, IP's
- Notation and Language
- Convex functions/sets, linear (in)equalities, LP geometry

SKILLS:

- Construct an MIP representing a given word problem
- Construct a constraint(s) from a logical description
- Compose a concise proof of a statement.
- Identify the features of an MIP
- Sketch the feasible region of an optimization problem.
- Interpret mathematical solutions of MIP's.

I will provide:

- Exam pages with room to write
- Any necessary Definitions / Theorems

You need to bring:

- you
- writing implements

Not Allowed:

- Any other materials ← that's right. No notes, book, calculator, paper, etc.
- Questions ← that's right. You are not allowed to ask any questions. If there seems to be an ambiguity then state your understanding, make a reasonable assumption, continue.